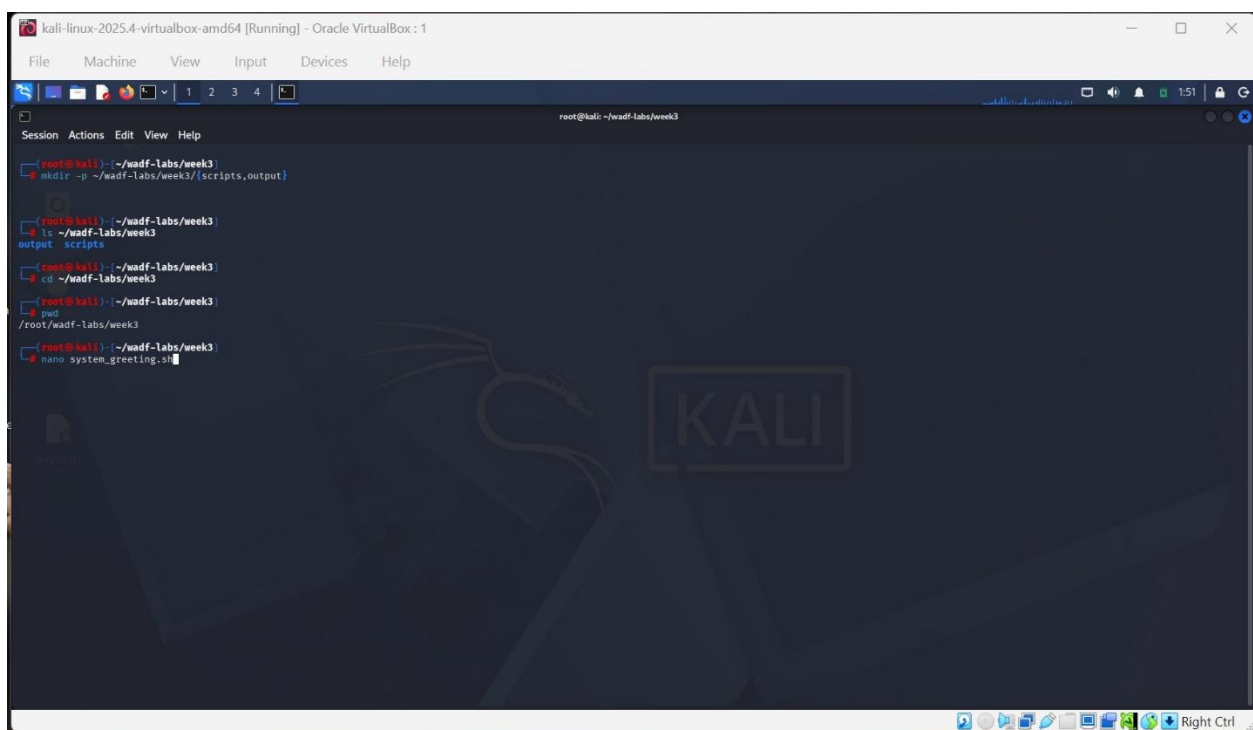


Paractical Lab

Name Maaz khan

5A



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help

root@kali: ~/wadf-labs/week3
Session Actions Edit View Help

root@kali) ~-/wadf-labs/week3
# mkdir -p ~/wadf-labs/week3/{scripts,output}

root@kali) ~-/wadf-labs/week3
# ls ~/wadf-labs/week3
output  scripts

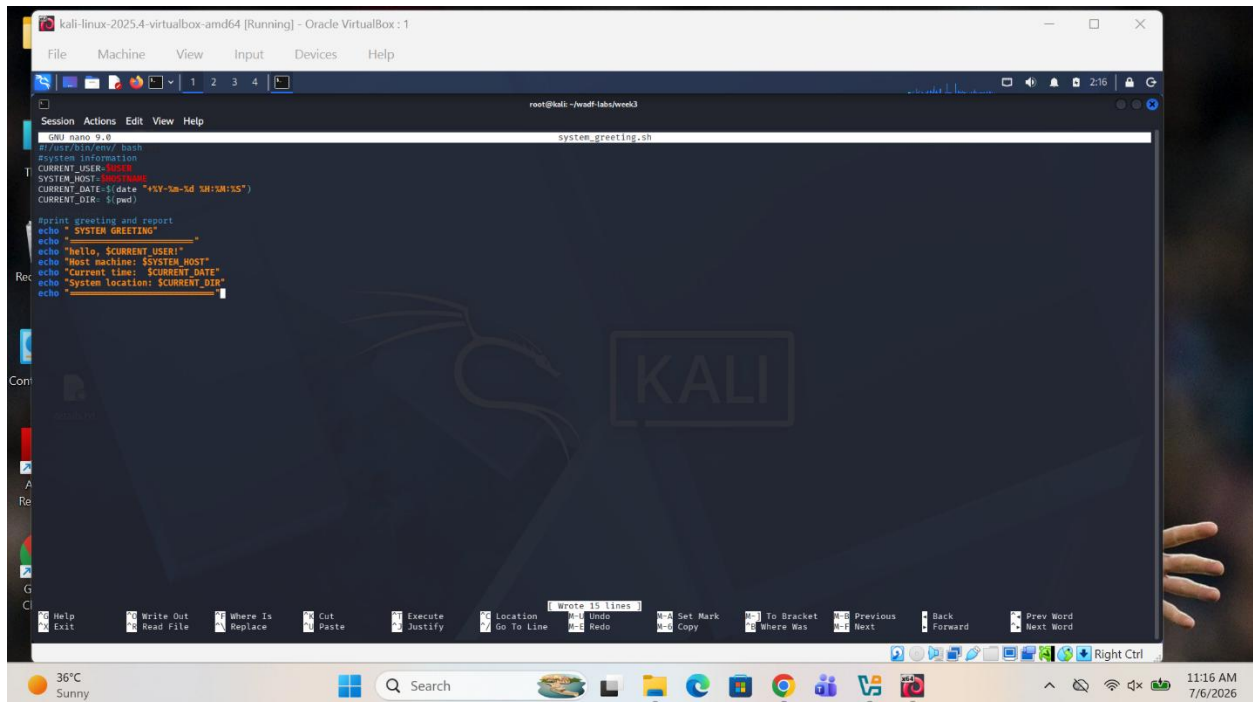
root@kali) ~-/wadf-labs/week3
# cd ~/wadf-labs/week3

root@kali) ~-/wadf-labs/week3
# pwd
/root/wadf-labs/week3

root@kali) ~-/wadf-labs/week3
# nano system_greeting.sh
```

I have done 3 tasks here , First I have make scripts and output directories in week 3 directory. Then I have entered into week 3

and use nano command. I have verify it through ls and pwd commands.



```
root@kali:~/work/week3# nano system_greeting.sh
# nano 9.0
# /usr/bin/env: bash
# system information
CURRENT_USER=$(whoami)
SYSTEM_HOST=$(hostname)
CURRENT_DATE=$(date +%Y-%m-%d %H:%M:%S)
CURRENT_DIR=$(pwd)

# print greeting and report
echo "SYSTEM GREETING"
echo "-----"
echo "Hello, $CURRENT_USER!"
echo "Host machine: $SYSTEM_HOST"
echo "Current time: $CURRENT_DATE"
echo "System location: $CURRENT_DIR"
echo "-----"
```

I have opened terminal text editor through nano command and making file and adding content for scripting.

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help

root@kali: ~/wadf-labs/week3/scripts
# cat system_greeting.sh
1 #!/usr/bin/env bash
2 # System Information
3 CURRENT_USER=$(whoami)
4 SYSTEM_HOST=$(hostname)
5 CURRENT_DATE=$(date "+%Y-%m-%d %H:%M:%S")
6 CURRENT_DIR=$(pwd)
7
8 # Print greeting and report
9 echo " SYSTEM GREETING"
10 echo "-----"
11 echo "Hello, $CURRENT_USER!"
12 echo "Host machine: $SYSTEM_HOST"
13 echo "Current time: $CURRENT_DATE"
14 echo "System location: $CURRENT_DIR"
15 echo "-----"

root@kali: ~/wadf-labs/week3/scripts
# chmod +x system_greeting.sh
root@kali: ~/wadf-labs/week3/scripts
# ./system_greeting.sh
SYSTEM GREETING
-----
Hello, root!
Host machine: kali
Current time: 2026-07-06 02:17:17
System location: /root/wadf-labs/week3/scripts
-----
```

First I have check file content here then I have checked its permissions and then grant executable permission and In last I have verified it again.

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help

root@kali: ~/wadf-labs/week3/scripts
# ./system_greeting.sh > ~/wadf-labs/week3/output/greeting_report.txt
system_greeting.sh: line 6: /root/wadf-labs/week3/scripts: is a directory
SYSTEM GREETING
-----
Hello, root!
Host machine: kali
Current time: 2026-07-06 02:19:27
System location: /root/wadf-labs/week3/scripts
-----

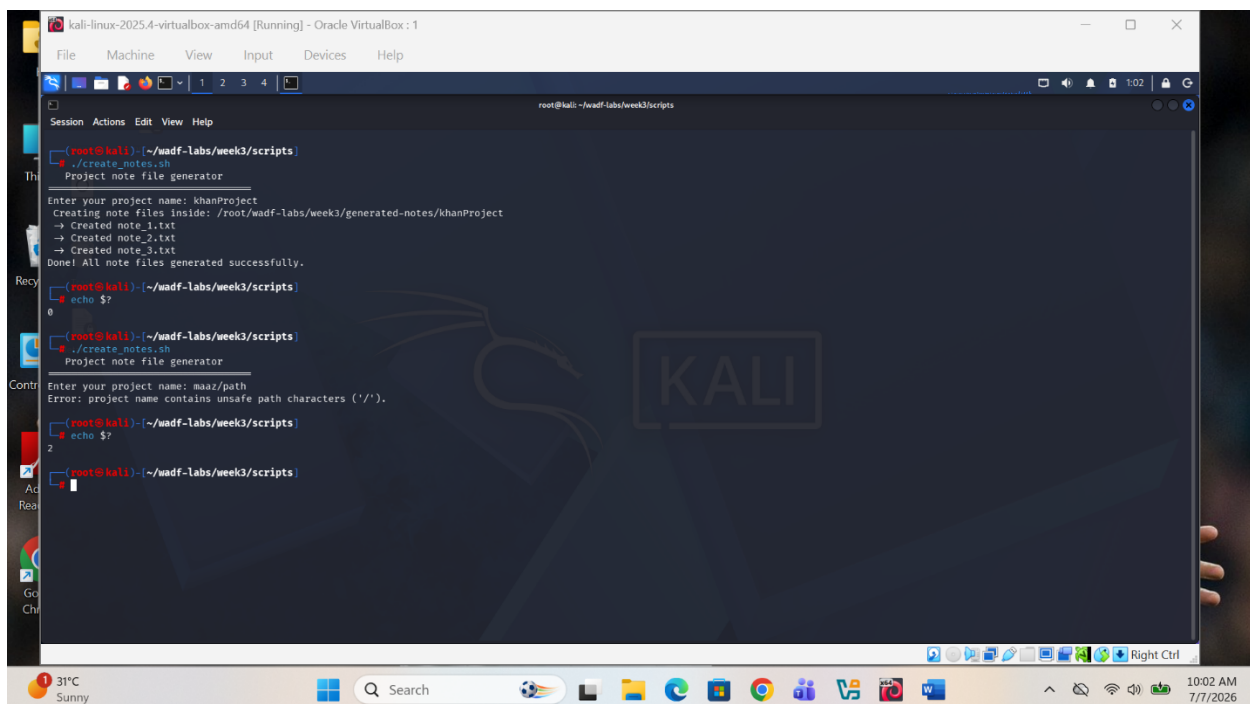
root@kali: ~/wadf-labs/week3/output
# ls
greeting_report.txt

root@kali: ~/wadf-labs/week3/output
# cat greeting_report.txt
SYSTEM GREETING
-----
Hello, root!
Host machine: kali
Current time: 2026-07-06 02:19:27
System location: /root/wadf-labs/week3/scripts
-----
```

I have run .sh file and save its output in txt file using tee pipeline. Then I have verified it through cat command.

5b

I have uses read -r to capture input, handles validation conditions if the file name is empty or unsafe path then it will through error and I have also use loops to generate the required project notes files.



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox: 1
File Machine View Input Devices Help

root@kali: ~/wadf-labs/week3/scripts
Session Actions Edit View Help

root@kali) ~/wadf-labs/week3/scripts
# ./create_notes.sh
Project note file generator

Enter your project name: khanProject
Creating note files inside: /root/wadf-labs/week3/generated-notes/khanProject
→ Created note_1.txt
→ Created note_2.txt
→ Created note_3.txt
Done! All note files generated successfully.

root@kali) ~/wadf-labs/week3/scripts
# echo $?
0

root@kali) ~/wadf-labs/week3/scripts
# ./create_notes.sh
Project note file generator

Enter your project name: maa2/path
Error: project name contains unsafe path characters ('/').

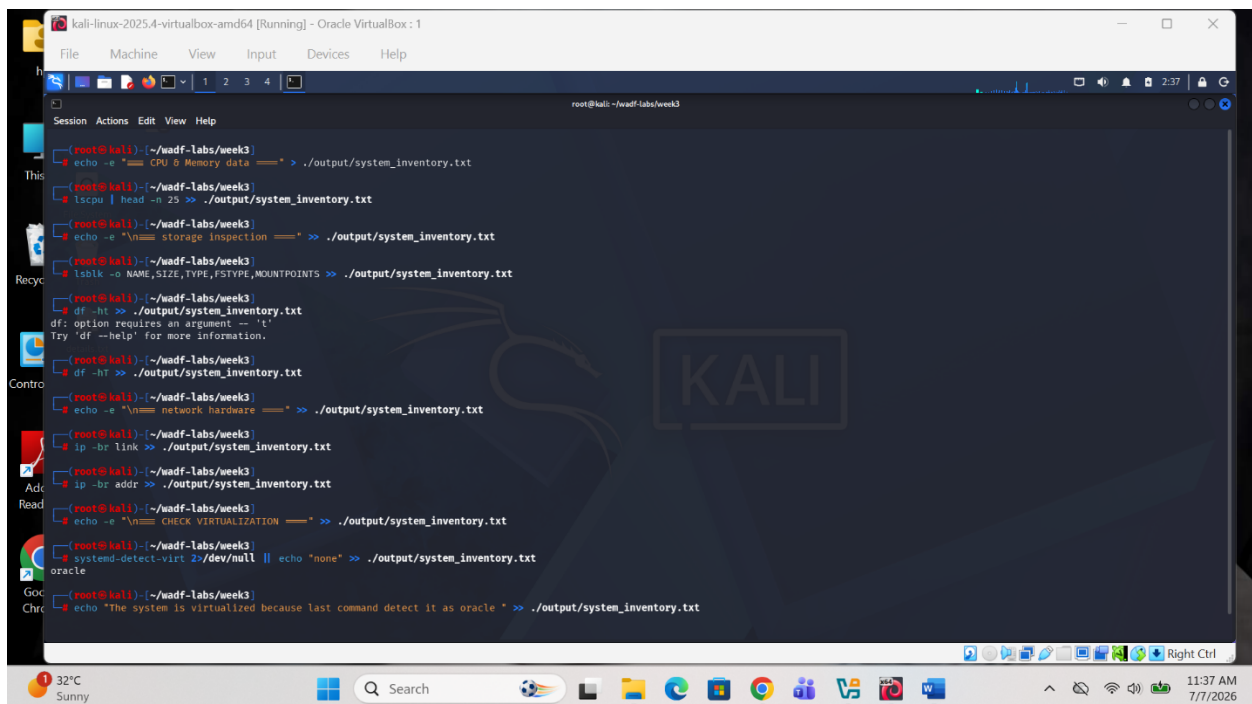
root@kali) ~/wadf-labs/week3/scripts
# echo $?
2

root@kali) ~/wadf-labs/week3/scripts
#
```

I have run the bash file and make dir there with khanProject name and the file were also created , I also verify it through echo \$? Command which gives 0 output it means the command is successful and I

also intentionally give wrong name using path name which gives me error .

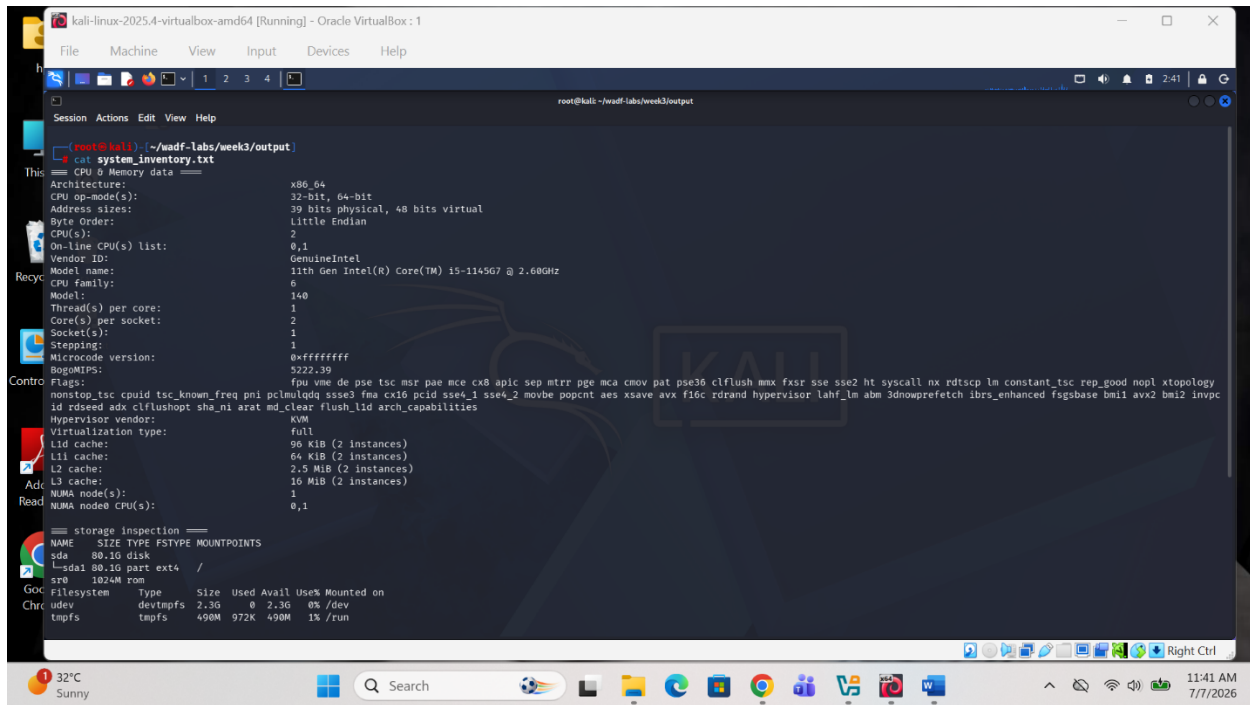
5c



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox: 1
File Machine View Input Devices Help

root@kali: ~/wadf-labs/week3
Session Actions Edit View Help
root@kali:~/wadf-labs/week3
root@kali:~/wadf-labs/week3# echo -e "=== CPU & Memory data ===" > ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# lscpu | head -n 25 >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# echo -e "\n=== storage inspection ===" >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# df -ht >> ./output/system_inventory.txt
df: option requires an argument -- 't'
Try 'df --help' for more information.
root@kali:~/wadf-labs/week3# df -ht >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# echo -e "\n=== network hardware ===" >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# ip -br link >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# ip -br addr >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# echo -e "\n=== CHECK VIRTUALIZATION ===" >> ./output/system_inventory.txt
root@kali:~/wadf-labs/week3# systemd-detect-virt 2 /dev/null || echo "none" >> ./output/system_inventory.txt
oracle
root@kali:~/wadf-labs/week3# echo "The system is virtualized because last command detect it as oracle" >> ./output/system_inventory.txt
```

I have collect cpu,memory,storage and network data and add it to inventory txt file.



The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox:1". The terminal is running the command `cat system_inventory.txt` in the directory `~/wadf-labs/week3/output`. The output of the command is a detailed system information report. The report includes sections for CPU & Memory data, storage inspection, and a table of filesystems.

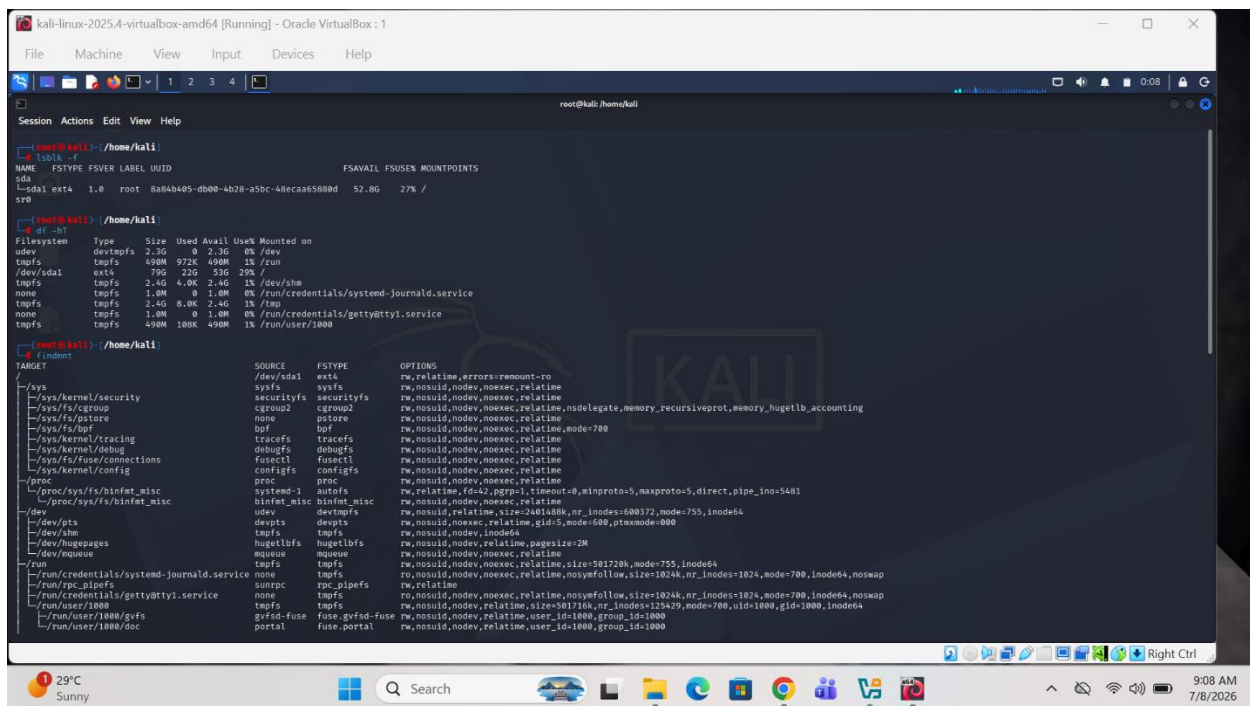
```
root@kali: ~/wadf-labs/week3/output
root@kali:~# cat system_inventory.txt

===== CPU & Memory data =====
Architecture: x86_64
CPU op-model(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: 11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz
CPU family: 6
Model: 140
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
stepping: 1
Microcode version: 0xffffffff
BogoMIPS: 5222.39
Flags: fpu vme de pse tsc mtr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ibrs_enhanced fsgsbase bti1 avx2 bmi1 invpc
id rsseed adx cflushopt sha_ni arat md_clear flush_lid arch_capabilities
Hypervisor vendor: KVM
Virtualization type: full
L1d cache: 96 KiB (2 instances)
L1i cache: 64 KiB (2 instances)
L2 cache: 2.5 MiB (2 instances)
L3 cache: 16 MiB (2 instances)
NUMA node(s): 1
NUMA node0 CPU(s): 0,1

===== storage inspection =====
NAME        SIZE TYPE FSTYPE MOUNTPOINTS
sda         80.1G disk /
l-sda1      80.1G part ext4 /
sr0         1024M rom
GOC Filesystem Type Size Used Avail Use% Mounted on
Chr: udev devtmpfs 2.3G 0 2.3G 0% /dev
tmpfs tmpfs 490M 972K 490M 1% /run
```

In last I have inspect system_inventory.txt file and verify that all information is add correctly .

6A

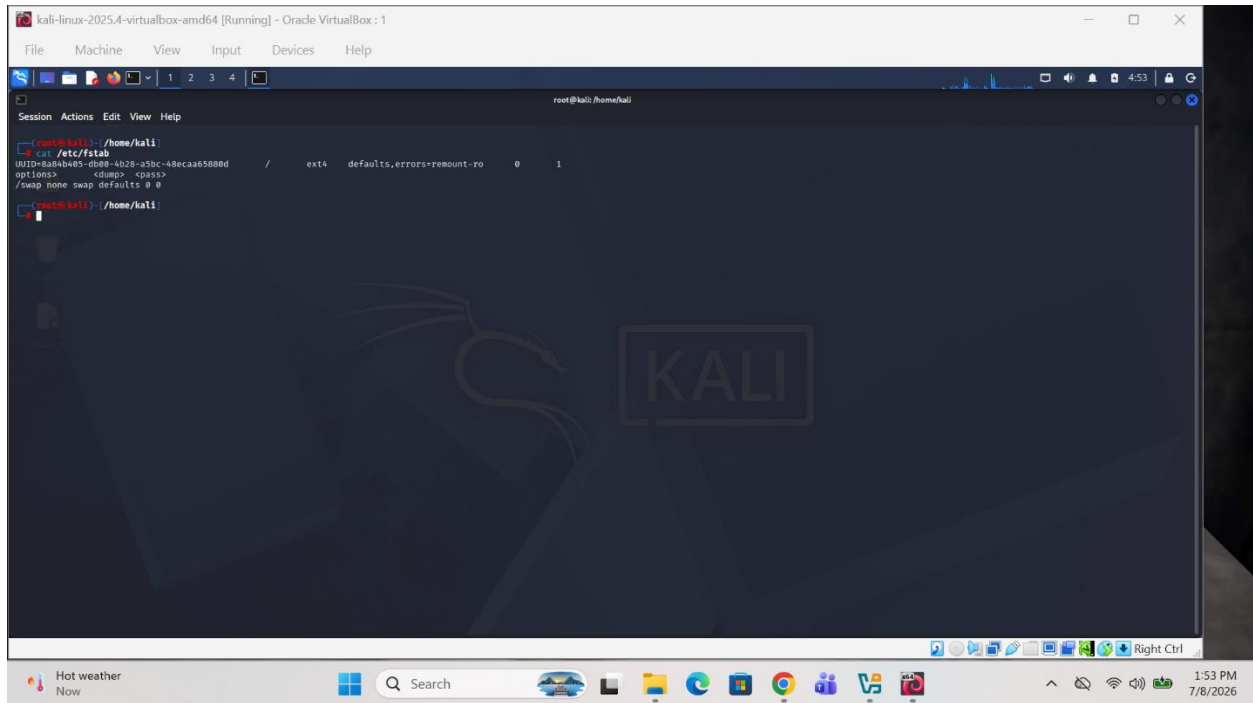


```
root@kali: /home/kali
lsblk -f
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSEN MOUNTPONTS
sda          ext4   1.0   root  8a84b405-d800-4b28-a5bc-48eca955880d  57.8G   27% /
sro          sro

root@kali: /home/kali
df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
udev            devtmpfs  2.3G   0 2.3G   0% /dev
tmpfs           tmpfs     490M  972K 490M   1% /run
/dev/sda1       ext4      79G   22G  53G   28% /
tmpfs           tmpfs     2.4G  4.0K  2.4G   1% /dev/shm
none            tmpfs     1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     2.4G  8.0K  2.4G   1% /tmp
none            tmpfs     1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     490M  100K 490M   1% /run/user/1000

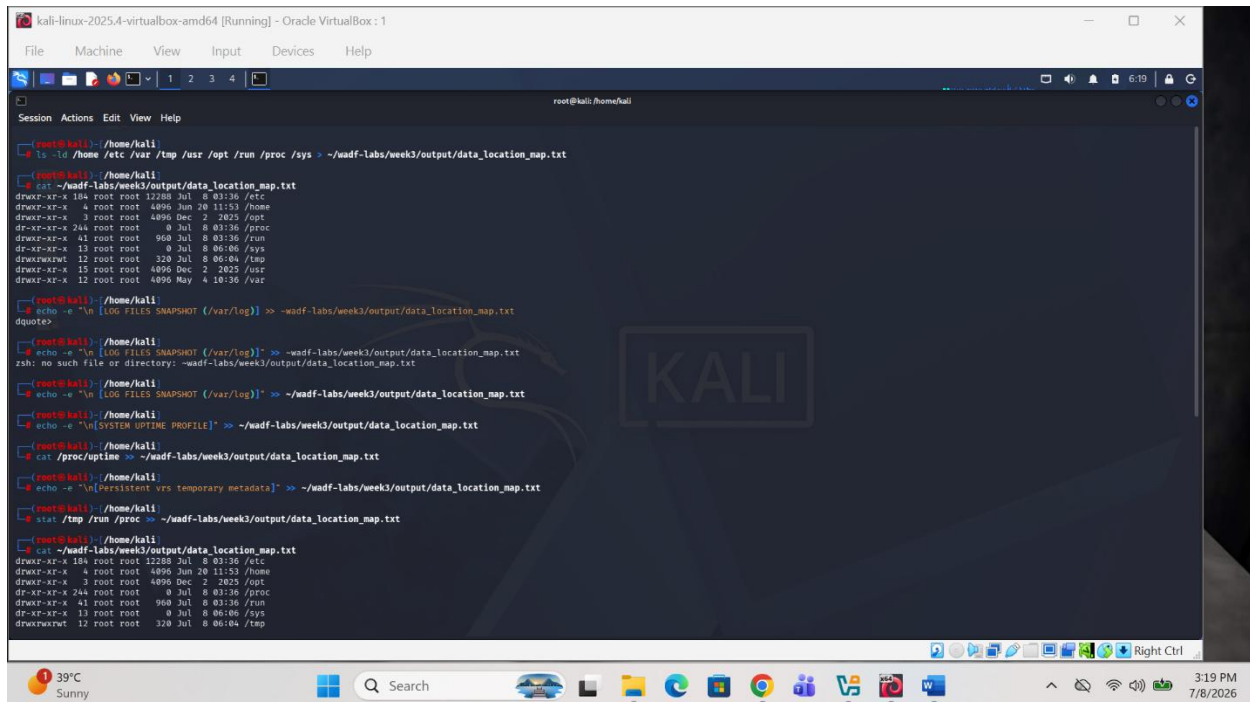
root@kali: /home/kali
findmnt
TARGET SOURCE FSTYPE OPTIONS
/sys /dev/sda1 ext4 rw,relatime,errors=remount-ro
/sys/kernel/security sysfs sysfs rw,nosuid,nodev,noexec,relatime
/sys/fs/cgroup cgroup2 cgroup2 rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot,memory_hugetlb_accounting
/sys/fs/pstore none pstore rw,nosuid,nodev,noexec,relatime
/sys/fs/bpf bpf bpf rw,nosuid,nodev,noexec,relatime,mode=700
/sys/kernel/tracing tracefs tracefs rw,nosuid,nodev,noexec,relatime
/sys/kernel/debug debugfs debugfs rw,nosuid,nodev,noexec,relatime
/sys/fs/fuse/connections fusectl fusectl rw,nosuid,nodev,noexec,relatime
/sys/kernel/config configfs configfs rw,nosuid,nodev,noexec,relatime
/proc proc proc rw,relatime,fd=42,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=5481
/proc/sys/fs/binfmt_misc binfmt_misc binfmt_misc rw,nosuid,nodev,noexec,relatime
/dev devtmpfs devtmpfs rw,nosuid,relatime,size=2481488,nr_inodes=608172,mode=755,inode64
/dev/pts devpts devpts rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000
/dev/shm tmpfs tmpfs rw,nosuid,nodev,inode64
/dev/hugepages hugetlbfs hugetlbfs rw,nosuid,nodev,relatime,pagesize=2M
/dev/mqueue mqueue mqueue rw,nosuid,nodev,noexec,relatime
/run runtmp tmpfs tmpfs rw,nosuid,nodev,noexec,relatime,size=581720k,mode=755,inode64
/run/credentials/systemd-journald.service none tmpfs rw,nosuid,nodev,noexec,relatime,nosymlinkfollow,size=1024k,nr_inodes=1024,mode=700,inode64,noswap
/run/rpc_pipefs none rpc_pipefs rw,relatime
/run/credentials/getty@tty1.service none tmpfs rw,nosuid,nodev,noexec,relatime,size=501716k,nr_inodes=12429,mode=700,uid=1000,gid=1000,inode64
/run/user/1000 gvfsd-fuse fuse.gvfsd-fuse rw,nosuid,nodev,relatime,user_id=1000,group_id=1000
/run/user/1000/doc portal fuse.portal rw,nosuid,nodev,relatime,user_id=1000,group_id=1000
```

I have use lsblk -f , df -hT and findmnt commands . it has given me block device and its partition . it has also show the total size ,used space, and filesystem type.



This command has shown how the system is configured to mount storage.

6B



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox: 1
File Machine View Input Devices Help

root@kali: /home/kali

Session Actions Edit View Help

root@kali: /home/kali
# ls -ld /home /etc /var /usr /opt /run /proc /sys > ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# cat ~/wadf-labs/week3/output/data_location_map.txt
drwxr-xr-x 184 root root 12288 Jul 8 03:36 /etc
drwxr-xr-x 4 root root 4096 Jun 20 11:53 /home
drwxr-xr-x 3 root root 4096 Dec 2 2025 /opt
dr-xr-xr-x 244 root root 0 Jul 8 03:36 /proc
drwxr-xr-x 41 root root 960 Jul 8 03:36 /run
dr-xr-xr-x 12 root root 0 Jul 8 06:06 /sys
drwxrwxrwt 12 root root 320 Jul 8 06:04 /tmp
drwxr-xr-x 15 root root 4096 Dec 2 2025 /usr
drwxr-xr-x 12 root root 4096 May 4 16:36 /var

root@kali: /home/kali
# echo -e "\n [LOG FILES SNAPSHOT (/var/log)]" >> ~/wadf-labs/week3/output/data_location_map.txt
dquote>

root@kali: /home/kali
# echo -e "\n [LOG FILES SNAPSHOT (/var/log)]" >> ~/wadf-labs/week3/output/data_location_map.txt
zsh: no such file or directory: ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# echo -e "\n [LOG FILES SNAPSHOT (/var/log)]" >> ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# echo -e "\n [SYSTEM UPTIME PROFILE]" >> ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# cat /proc/uptime >> ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# echo -e "\n [persistent vrs temporary metadata]" >> ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# stat /tmp /run /proc >> ~/wadf-labs/week3/output/data_location_map.txt

root@kali: /home/kali
# cat ~/wadf-labs/week3/output/data_location_map.txt
drwxr-xr-x 184 root root 12288 Jul 8 03:36 /etc
drwxr-xr-x 4 root root 4096 Jun 20 11:53 /home
drwxr-xr-x 3 root root 4096 Dec 2 2025 /opt
dr-xr-xr-x 244 root root 0 Jul 8 03:36 /proc
drwxr-xr-x 41 root root 960 Jul 8 03:36 /run
dr-xr-xr-x 12 root root 0 Jul 8 06:06 /sys
drwxrwxrwt 12 root root 320 Jul 8 06:04 /tmp
```

First I have add record the ownership of files in data_location_map.txt. then I have used find and cat command to inspect local logs and stream system information data cleanly and add it same file. In last I have use stat command for persistent vrs temporary memory status and add its output to file.

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox: 1
File Machine View Input Devices Help

root@kali: /home/kali
Session Actions Edit View Help

root@kali: /home/kali
cat ~/hackt-lab/week3/output/data_location_map.txt
drwxr-xr-x 184 root root 12288 Jul  8 03:36 /etc
drwxr-xr-x  4 root root 4096 Jun 20 11:53 /home
drwxr-xr-x  3 root root 4096 Dec  2 2025 /opt
dr-xr-xr-x 244 root root  0 Jul  8 03:36 /proc
drwxr-xr-x 41 root root 960 Jul  8 03:36 /run
dr-xr-xr-x 13 root root  0 Jul  8 06:06 /sys
drwxrwxrwt 12 root root 320 Jul  8 06:04 /tmp
drwxr-xr-x 15 root root 4096 Dec  2 2025 /usr
drwxr-xr-x 12 root root 4096 May  4 10:36 /var

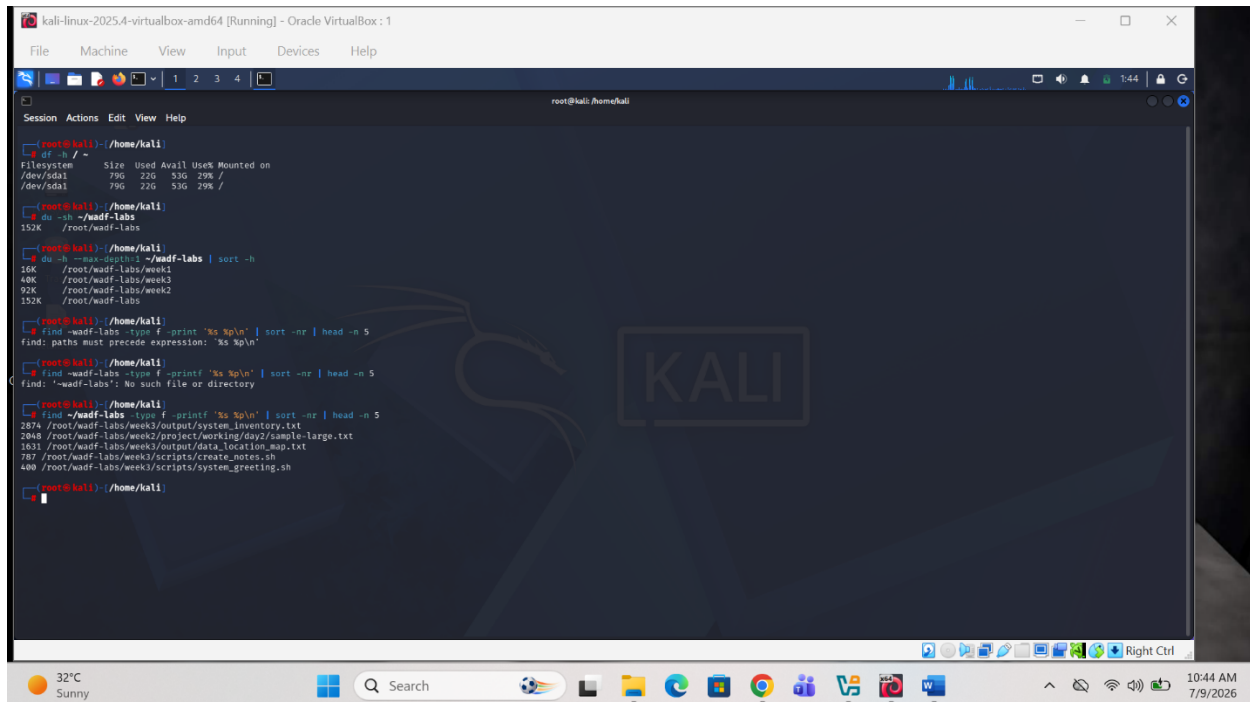
[LOG FILES SNAPSHOT (/var/log)]

[SYSTEM UPTIME PROFILE]
9493.77 16988.26

[Persistent vrs temporary metadata]
File: /tmp
Size: 320      Blocks: 0      IO Block: 4096  directory
Device: 0,34  Inode: 1      Links: 12
Access: (1777/drwxrwxrwt)  Uid: (  0/   root)  Gid: (  0/   root)
Change: 2026-07-08 03:36:30.652000037 -0400
Modify: 2026-07-08 06:09:20.335092191 -0400
Birth: 2026-07-08 03:36:30.652000037 -0400
File: /run
Size: 960      Blocks: 0      IO Block: 4096  directory
Device: 0,24  Inode: 1      Links: 41
Access: (0755/drwxr-xr-x)  Uid: (  0/   root)  Gid: (  0/   root)
Change: 2026-07-08 03:36:25.888000008 -0400
Modify: 2026-07-08 03:36:59.212599686 -0400
Birth: 2026-07-08 03:36:59.212599686 -0400
File: /proc
Size: 0        Blocks: 0      IO Block: 1024  directory
Device: 0,22  Inode: 1      Links: 238
Access: (0555/dr-xr-xr-x)  Uid: (  0/   root)  Gid: (  0/   root)
Change: 2026-07-08 03:36:25.888000008 -0400
Modify: 2026-07-08 03:36:25.888000008 -0400
Birth: -
root@kali: /home/kali
```

I have use cat command to inspect a file and verified that all data has successfully added in file.

6C



```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox: 1
File Machine View Input Devices Help

root@kali: /home/kali
root@kali: /home/kali
df -h / ~
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        79G   22G   53G   29% /
/dev/sda1        79G   22G   53G   29% /

root@kali: /home/kali
du -sh ~/wadf-labs
152K    /root/wadf-labs

root@kali: /home/kali
du -sh --max-depth=1 ~/wadf-labs | sort -h
16K     /root/wadf-labs/week1
48K     /root/wadf-labs/week3
92K     /root/wadf-labs/week2
152K    /root/wadf-labs

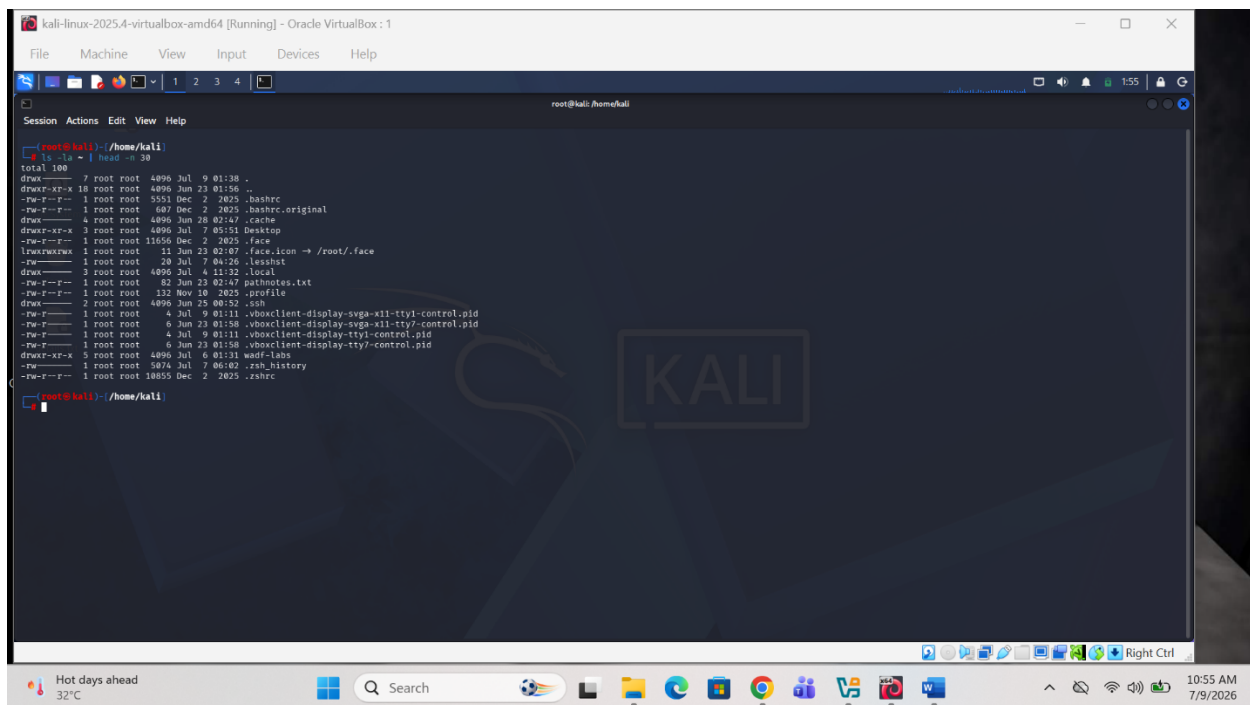
root@kali: /home/kali
find ~/wadf-labs -type f -print '%s %p\n' | sort -nr | head -n 5
find: paths must precede expression: '%s %p\n'

root@kali: /home/kali
find ~/wadf-labs -type f -printf '%s %p\n' | sort -nr | head -n 5
find: ~/wadf-labs: No such file or directory

root@kali: /home/kali
find ~/wadf-labs -type f -printf '%s %p\n' | sort -nr | head -n 5
2876 /root/wadf-labs/week3/output/system_inventory.txt
2040 /root/wadf-labs/week2/project/working/day2/sample-large.txt
1631 /root/wadf-labs/week3/output/data_location_map.txt
287 /root/wadf-labs/week3/scripts/create_notes.sh
400 /root/wadf-labs/week3/scripts/system_greeting.sh

root@kali: /home/kali
```

I have use some commands here. First I have check the over all capacity and available space of my root (/) and home (~) storage. Then I have check total size of my workspace lab and then check the space of subdirectories in it and in last I have locate and list top 5 largest individual files inside my lab workspace and print it in bytes along with their path.



```
root@kali: ~/home/kali
ls -la
total 100
drwx----- 7 root root 4096 Jul 9 01:38 .
drwxr-xr-x 18 root root 4096 Jun 23 01:56 ..
-rw-r--r-- 1 root root 5551 Dec 2 2025 .bashrc
-rw-r--r-- 1 root root 687 Dec 2 2025 .bashrc.original
drwx----- 4 root root 4096 Jun 28 02:47 .cache
drwxr-xr-x 3 root root 4096 Jul 7 05:51 desktop
-rw-r--r-- 1 root root 11656 Dec 2 2025 .face
lrwxrwxrwx 1 root root 11 Jun 23 02:07 .face.icon -> /root/.face
-rw----- 1 root root 20 Jul 7 06:26 .lesshst
drwx----- 3 root root 4096 Jul 4 11:32 .local
-rw-r--r-- 1 root root 82 Jun 23 02:47 .pathnotes.txt
-rw-r--r-- 1 root root 132 Nov 10 2025 .profile
drwx----- 2 root root 4096 Jun 25 08:52 .ssh
-rw-r--r-- 1 root root 4 Jul 9 01:11 .vboxclient-display-svga-all-tty-control.pid
-rw-r--r-- 1 root root 6 Jun 23 01:58 .vboxclient-display-svga-all-tty-control.pid
-rw-r--r-- 1 root root 4 Jul 9 01:11 .vboxclient-display-tty-control.pid
-rw-r--r-- 1 root root 6 Jun 23 01:58 .vboxclient-display-tty-control.pid
drwxr-xr-x 5 root root 4096 Jul 6 01:31 wadf-labs
-rw----- 1 root root 5874 Jul 7 06:02 zsh_history
-rw-r--r-- 1 root root 18855 Dec 2 2025 .zshrc
```

I have checked my home directory for large hidden files or directories that don't show up in a standard listing.

Recommend Action :

The investigation shows that sidk space is ostly taken up by specific large log and temporary files inside the ~/wadf-labs directory . Rather than running a risky broad cleanup, it is recommended to selectively compress these inactive files into archives to safely reclaim space.